

[INCIDENT HANDLER JOURNAL]

Analyst: marshall_c

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[ENTRY #01: DOCUMENTING A CYBERSECURITY INCIDENT]

Date: May 3, 2026

Description: This entry serves as the formal ingestion and preliminary documentation of a critical cybersecurity incident impacting a primary care medical facility. The incident involves the unauthorized encryption of sensitive datasets and the complete cessation of clinical operations. This log initiates the "Identification" phase of the incident response life cycle, focusing on the correlation between spear phishing delivery and the subsequent ransomware payload execution.

Tools Used:

- **Endpoint Telemetry:** Utilized to identify the execution path of the malicious attachment and the resulting cryptographic activity.
- **Mail Gateway Logs:** Accessed to isolate the specific phishing emails and identify additional recipients within the organization who may be compromised.
- **Security Information and Event Management (SIEM):** Employed to visualize the timeline of the initial intrusion and identify lateral movement attempts across the clinic's network.

The 5 Ws:

- **Who:** An organized cybercriminal syndicate with a documented history of targeting high-value infrastructure within the healthcare and transportation sectors. Their sophisticated approach suggests a focus on sectors where downtime directly impacts life safety, increasing ransom leverage.
- **What:** A catastrophic Ransomware infection that has successfully encrypted critical patient medical records and Protected Health Information (PHI). This has resulted in a total "denial of service" for clinical staff, rendering internal software and scheduling systems completely inaccessible.
- **Where:** A small, U.S. based healthcare clinic specializing in primary care. The scope of the incident appears localized to the internal clinic network but involves external communication with threat actor-controlled Command and Control (C2) servers.

- **When:** The anomaly was first reported by staff at approximately 09:00 a.m. local time on Tuesday. Rapid encryption suggests the payload was likely executed shortly before or during the start of business hours to maximize operational impact.
- **Why:** The breach was facilitated by a targeted Spear Phishing campaign. Attackers utilized psychological manipulation to entice employees into downloading a malicious attachment. This bypassed existing perimeter defenses, allowing the ransomware to execute with the permissions of the local user and propagate through shared network directories.

Additional Notes:

1. **Backup Integrity Assessment:** While a ransom has been demanded, the primary unanswered question remains: Does the organization possess “Air-Gapped” or immutable backups that have remained untouched by the encryption process, or has the ransomware successfully compromised the backup server itself?
2. **Scope of Exfiltration:** Beyond the encryption of files, is there evidence of data exfiltration (Double Extortion)? We must determine if patient PHI was transmitted to external servers prior to the encryption phase, which would trigger mandatory HIPAA breach notification protocols.
3. **Endpoint Hardening Gaps:** Why did the existing antivirus or endpoint protection fail to flag the malicious attachment at the point of download? A thorough review of the “Execution” logs is required to determine if the malware utilized a Zero Day exploit or if security policies were improperly configured.
4. **Regulatory Obligations:** At what point will the clinic need to notify local law enforcement and federal agencies (CISA/FBI) given the organized nature of the threat actors and the sensitive nature of the encrypted data?

[ENTRY #02: ANALYZING A PACKET CAPTURE FILE]

Date: May 3, 2026

Description: This entry documents the technical evaluation and analysis of network traffic utilizing **Wireshark**. This investigation is situated within the **Detection and Analysis** phase of the NIST Incident Response Lifecycle. The primary objective was to decompose complex packet captures into human-readable protocol hierarchies to identify anomalous patterns, such as suspicious beaconing or unauthorized data transit, that characterize an active intrusion.

Tools Used:

- **Wireshark (GUI Protocol Analyzer):** Leveraged for deep packet inspection (DPI) and the reconstruction of TCP streams to visualize the bidirectional communication

between internal endpoints and external Command and Control (C2) infrastructure.

- **Capture Filters:** Applied to isolate specific protocols (e.g., HTTP, DNS, ICMP) to reduce noise and focus on high-fidelity indicators of malicious traffic.

Additional Notes:

The transition to a graphical analysis environment was critical for visualizing the “Expert Info” logs provided by Wireshark, which flags malformed packets and suspicious retransmissions. This tool is indispensable for moving beyond simple alert triage into a deep-dive forensic investigation where the specific mechanics of a network breach must be understood.

[ENTRY #03: CAPTURING MY FIRST PACKET]

Date: May 3, 2026

Description: This entry focuses on the tactical capture of live network traffic using **tcpdump** within a Command Line Interface (CLI) environment. This activity supports the **Detection and Analysis** phase, providing a lightweight and highly efficient method for gathering forensic evidence directly from the source. In high-stakes environments where system overhead must be minimized, tcpdump serves as the primary tool for securing raw packet data for later analysis.

Tools Used:

- **tcpdump (CLI Packet Sniffer):** Employed for the capture of raw frames and headers. The utility was configured to write output to .pcap files, ensuring that the integrity of the captured data was preserved for subsequent forensic review in a controlled environment.
- **CLI Flags:** Used to specify interface listening (e.g., -i eth0) and to limit capture size or count to prevent storage exhaustion during high-volume traffic events.

Additional Notes:

Mastering the CLI syntax remains a significant hurdle; however, the ability to script traffic captures and pipe them into other security utilities provides an operational speed that cannot be matched by GUI-based tools. This exercise highlighted the critical balance between technical precision in command entry and the necessity of rapid data collection during an ongoing incident.

[ENTRY #04: INVESTIGATE A SUSPICIOUS FILE HASH]

Date: May 3, 2026

Description: This entry details a targeted malware investigation initiated by a SOC alert

within a financial services infrastructure. The investigation is rooted in the **Detection and Analysis** phase, focusing on the static and behavioral verification of a suspicious file hash (54e6ea...). The goal was to pivot from a singular indicator (the hash) to a broader set of technical Indicators of Compromise (IoCs) to inform defensive posturing and containment strategies.

Tools Used:

- **VirusTotal (Threat Intelligence Aggregator):** Utilized to correlate the suspicious hash against multiple global antivirus engines and community reputation scores.
- **Pyramid of Pain Framework:** Applied to categorize and prioritize IoCs (Hashes, IPs, Domains) based on the level of “pain” or operational difficulty their mitigation causes the threat actor.

The 5 Ws:

- **Who:** An unknown external threat actor utilizing a password-protected spreadsheet to bypass standard mail gateway sandbox inspections.
- **What:** Investigation of a malicious payload with SHA-256 hash: 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b, which was identified as a known Trojan variant.
- **Where:** A local workstation within a financial services corporate network.
- **When:** 1:20 p.m. local time, following an Intrusion Detection System (IDS) alert generated by the creation of unauthorized executable files.
- **Why:** An employee was social-engineered into downloading an email attachment and entering a provided password, thereby authorizing the execution of the malicious payload which bypassed signature-based defenses.

[REFLECTIONS AND NOTES]

1. **Technical Challenges:** The most persistent challenge was achieving operational fluency with **tcpdump** command-line syntax. As an analyst transitioning into CLI environments, the precision required to apply complex filters without error was a steep learning curve. I encountered significant frustration when captures did not yield expected results, which ultimately reinforced the professional necessity of meticulous instruction-following and the verification of each command's output.
2. **Evolution of Understanding:** My perspective on incident detection and response has shifted from a series of reactive, isolated tasks to a holistic, lifecycle-driven methodology. I now prioritize the “Golden Thread” of evidence that connects an initial phishing email to a high-impact ransomware event. Understanding the **NIST Incident Response Lifecycle** has provided the structural framework needed to

manage complex security incidents with analytical composure rather than just clearing alerts.

3. **Professional Interests:** I found the most engagement in the study of **Network Traffic Analysis**. Using tools like Wireshark and tcpdump to “see” the invisible communication between systems was a transformative experience. This interest has developed into a professional goal to specialize in network forensics and threat hunting, as the ability to interpret raw packet data is the ultimate skill set for a high-tier security professional.